

# **Basic Concepts in Digital Image Processing**

# **Announcement**

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**Homework #1 was posted in Blackboard and on class website.**

**Due time: 1:15pm Tuesday, Feb 3**

**Submit it via Blackboard**

# **Today's Agenda**

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- **Interpolation**
- **Basic Relationships between Pixels**
- **Mathematical tools for image processing**

# Spatial Resolution

**Spatial resolution: smallest discernible details**

- # of line pairs per unit distance
- # of dots (pixels) per unit distance
  - Printing and publishing
  - In US, dots per inch (dpi)

Newspaper → magazines → book



**Large image size itself does not mean high spatial resolution!**

→ Scene/object size in the image



1280\*960



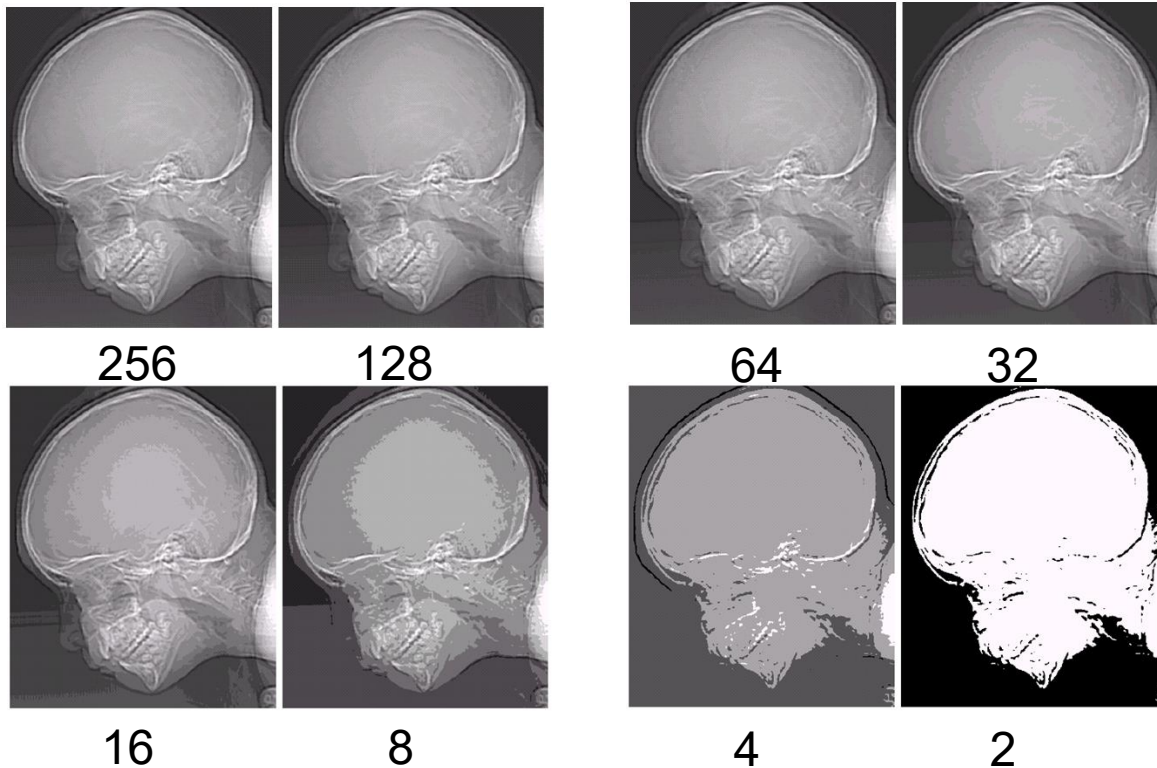
a b  
c d

**FIGURE 2.20** Typical effects of reducing spatial resolution. Images shown at: (a) 1250 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi. The thin black borders were added for clarity. They are not part of the data.

# Intensity Resolution

## Intensity resolution

- Smallest discernible change in intensity levels
- Using the number of levels of intensities
- False contouring (banding) when  $k$  is small - undersampling



# Isopreference Curves

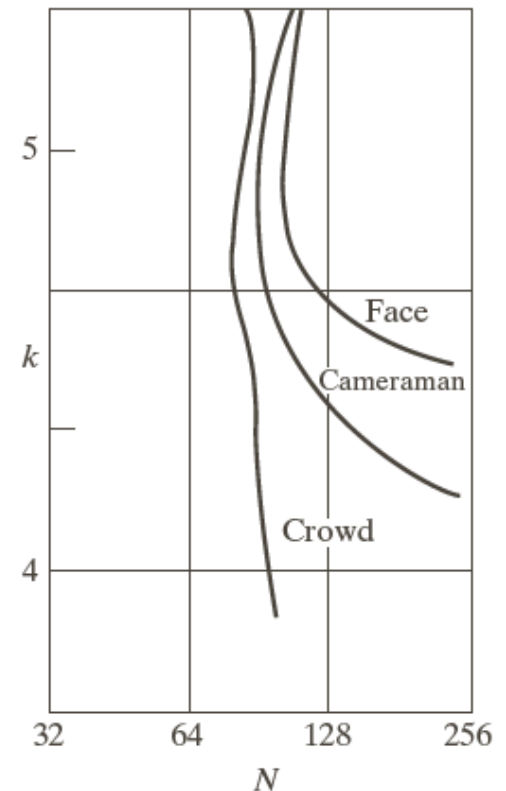


a b c

**FIGURE 2.22** (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail. (Image (b) courtesy of the Massachusetts Institute of Technology.)

**Vary the spatial and intensity sampling simultaneously:**

**FIGURE 2.23**  
Typical isopreference curves for the three types of images in Fig. 2.22.



# Image Resampling & Interpolation

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## Need to resample the image when

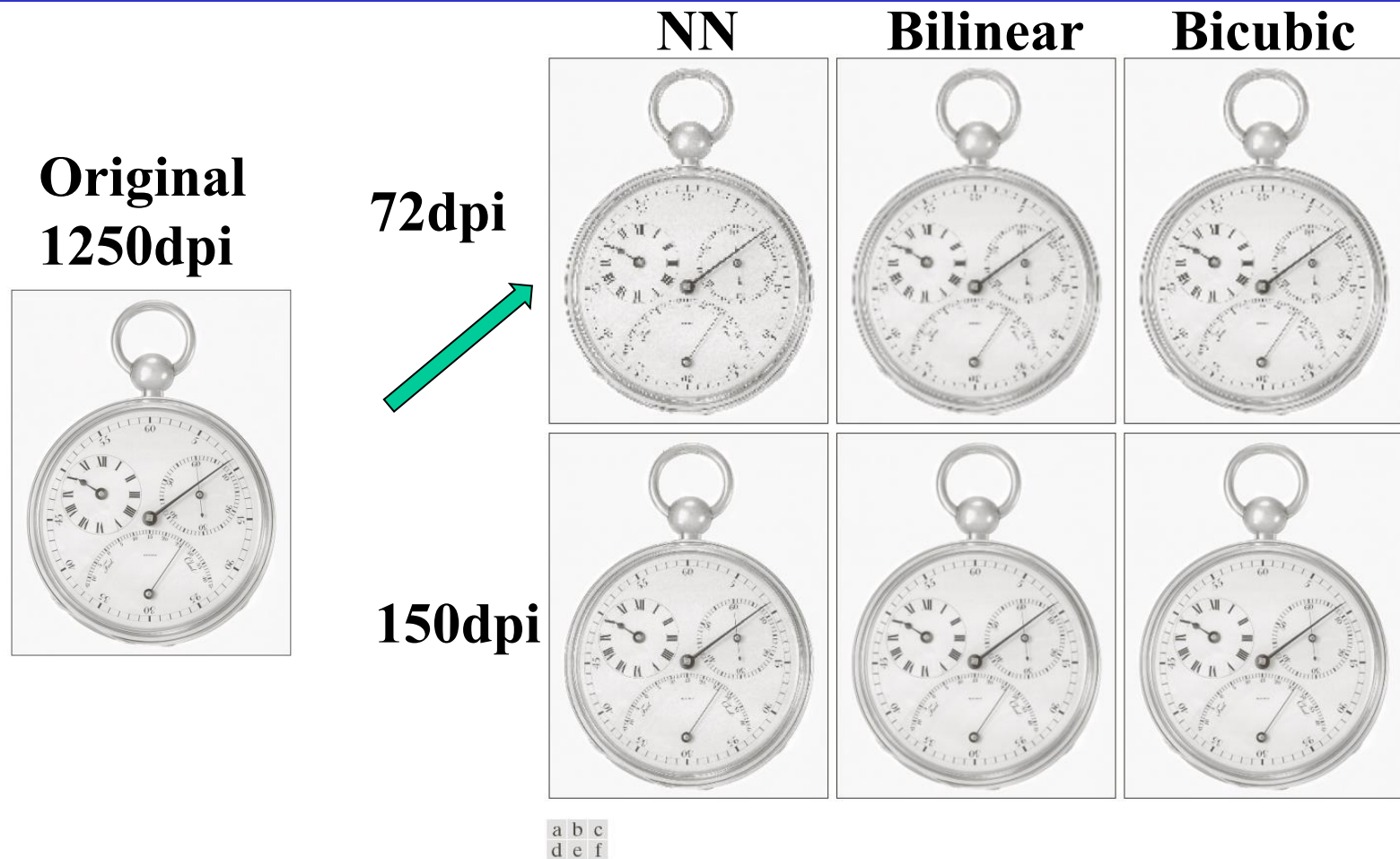
- Rescaling
- Geometrical transformation
- The output image coordinates are not discrete

## Interpolation methods:

- Nearest neighbor
  - Fast and simple
  - Loss of sharpness
  - Artifacts (checkerboard)
- Bilinear
- Bicubic
  - Images are sharpest
  - Fine details are preserved
  - Slow



# Image Resampling & Interpolation

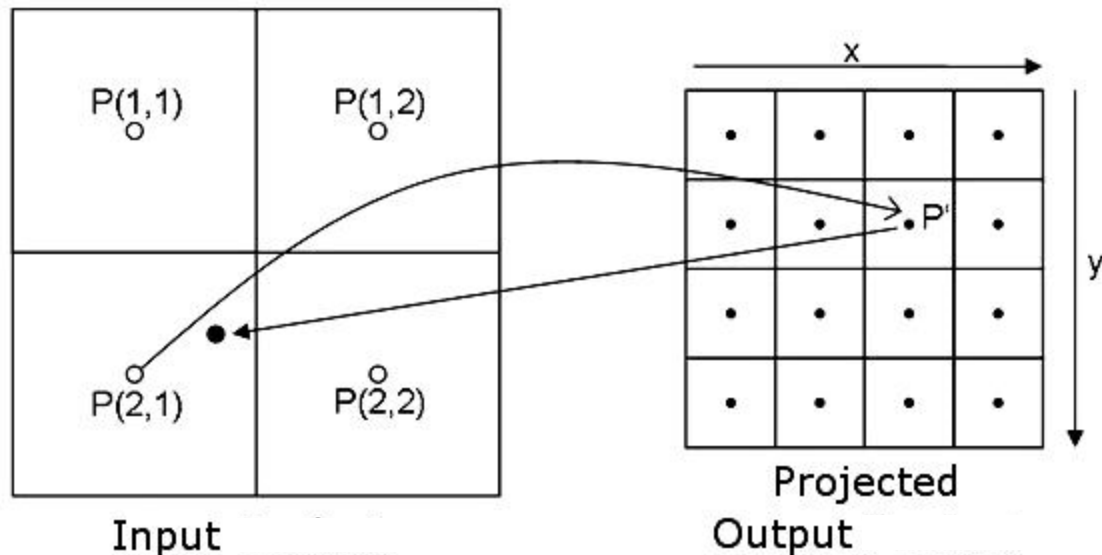


**FIGURE 2.24** (a) Image reduced to 72 dpi and zoomed back to its original size ( $3692 \times 2812$  pixels) using nearest neighbor interpolation. This figure is the same as Fig. 2.20(d). (b) Image shrunk and zoomed using bilinear interpolation. (c) Same as (b) but using bicubic interpolation. (d)–(f) Same sequence, but shrinking down to 150 dpi instead of 72 dpi [Fig. 2.24(d) is the same as Fig. 2.20(c)]. Compare Figs. 2.24(e) and (f), especially the latter, with the original image in Fig. 2.20(a).



# Image Interpolation – Nearest Neighbor

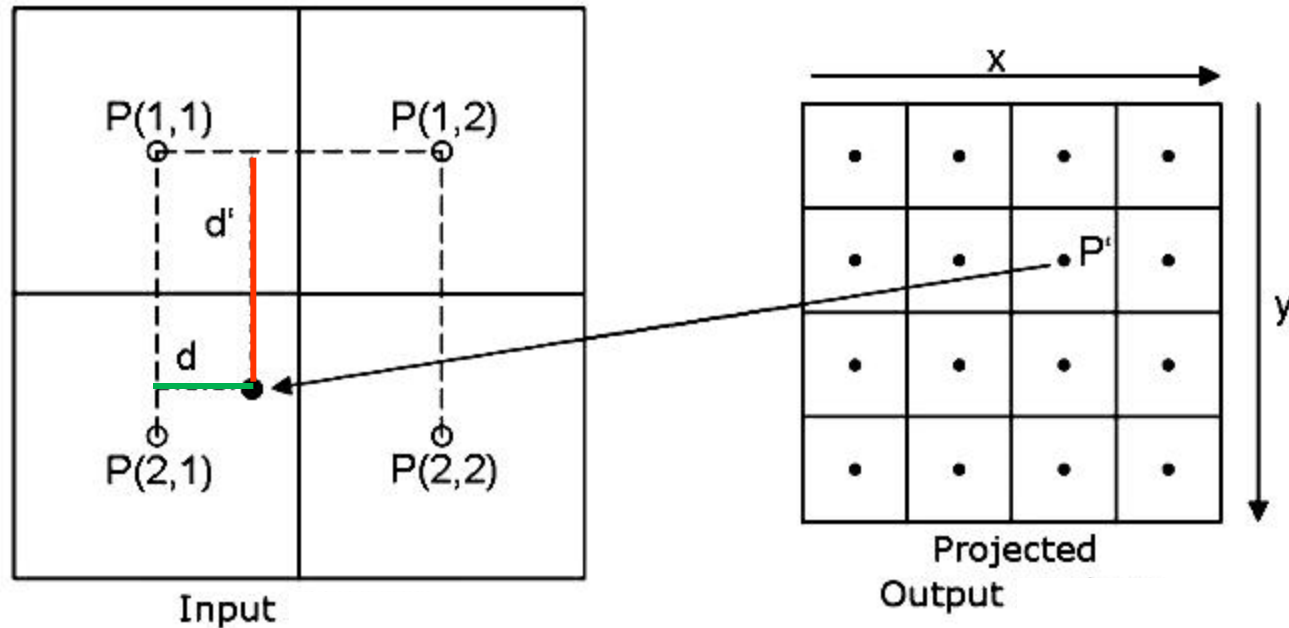
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<http://www.brockmann-consult.de/beam/doc/help/general/ResamplingMethods.html>

**Assign each pixel in the output image with the nearest neighbor in the input image.**

# Image Interpolation – Bilinear



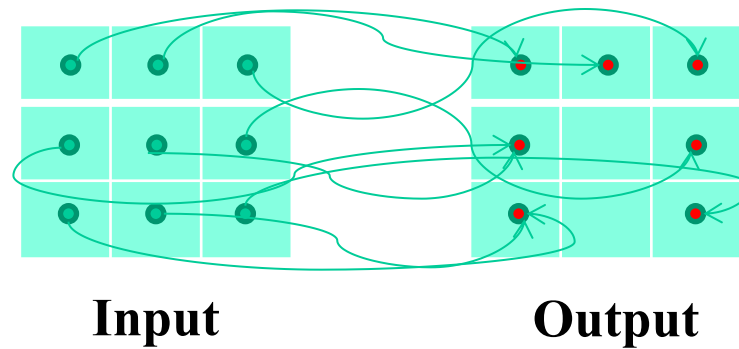
<http://www.brockmann-consult.de/beam/doc/help/general/ResamplingMethods.html>

$$\begin{aligned} P' = & P(1, 1)(1 - d)(1 - d') \\ & + P(1, 2)d(1 - d') + P(2, 1) * d' \\ & * (1 - d) + P(2, 2)dd' \end{aligned}$$

# Image Resampling & Interpolation

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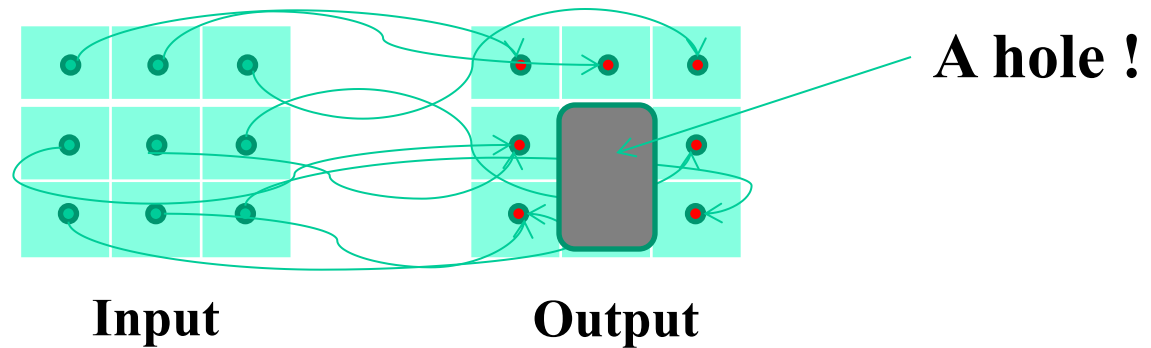
## Forward mapping



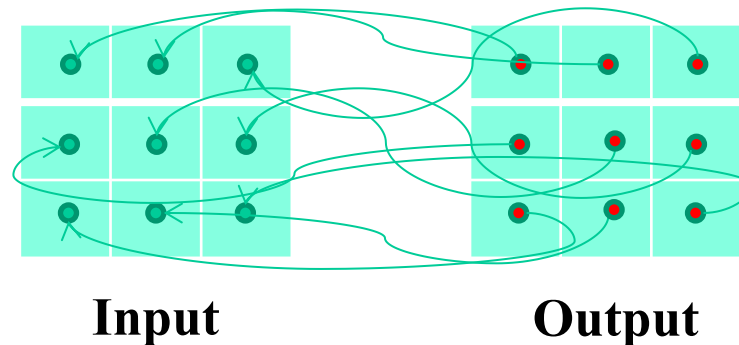
# Issues on Image Resampling & Interpolation

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## Missing points in forward mapping



## Solution: perform a backward mapping



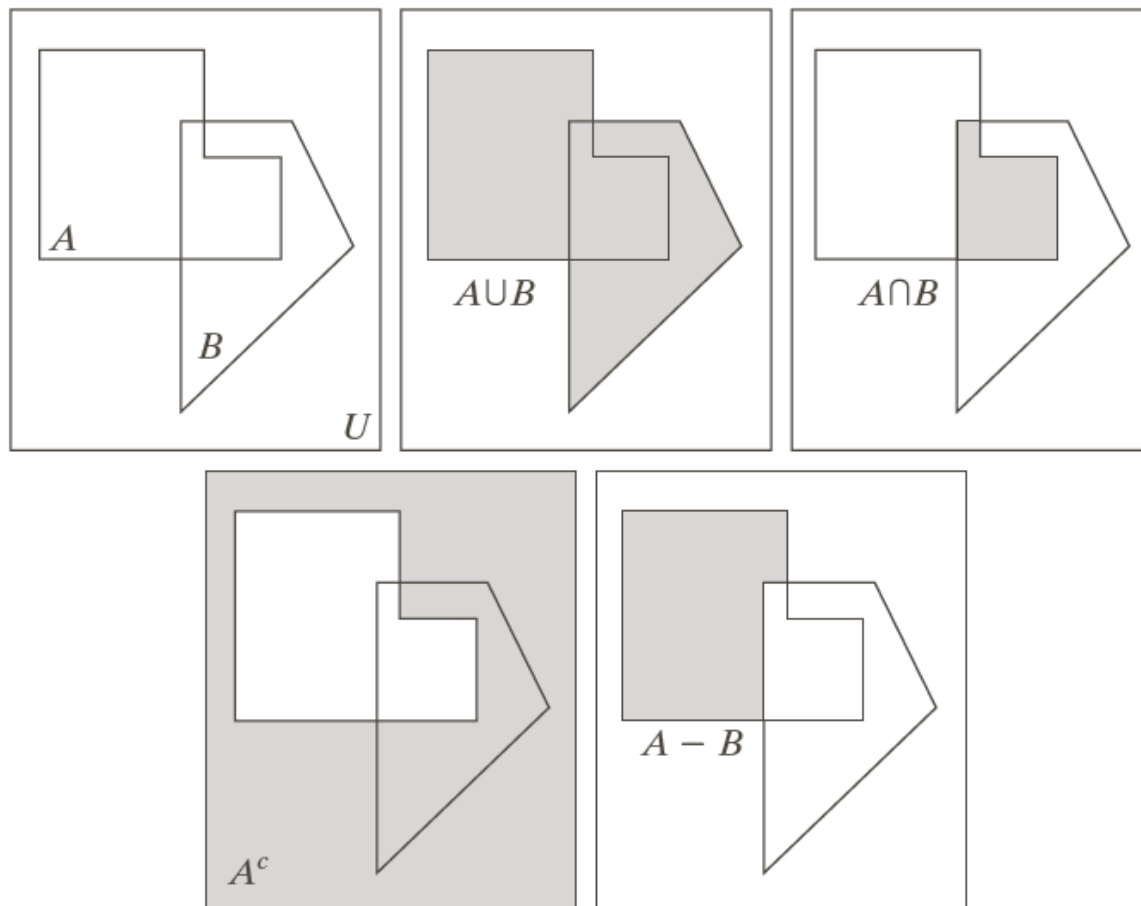
# Basic Set and Logical Operations

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- **A is a set:  $A=\{.\}$       e.g.  $A=\{1,...,255\}$  or  $A = \{w|w = 1, ..., 255\}$**   
 **$A = \emptyset$       for empty set**
- **a is an element of A ( $a \in A$ ) or a isn't an element of A ( $a \notin A$ )**
- **A is a *subset* of B if every element in A also is in B ( $A \subseteq B$ )**
- **C is the *union* of two sets A and B ( $C = A \cup B$ )**
- **C is the *intersection* of A and B ( $C = A \cap B$ )**
- **Disjoint or mutual exclusive sets ( $A \cap B = \emptyset$ )**
- **Set *universe* is the set of all elements in an application**
- **Set *difference* ( $A - B = \{w|w \in A, w \notin B\}$ )**

# Set Operations Based on Coordinates

A region in an image is represented by a set of coordinates within the region



|   |   |   |
|---|---|---|
| a | b | c |
| d | e |   |

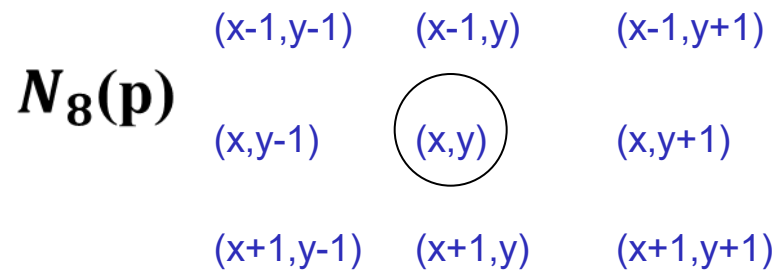
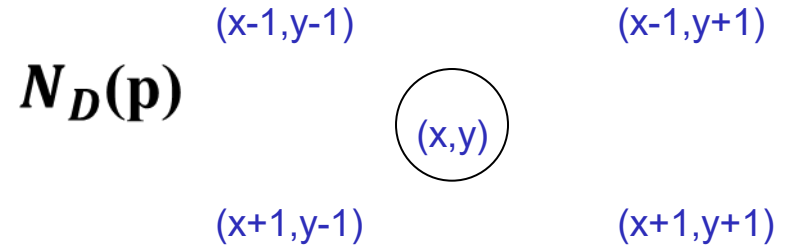
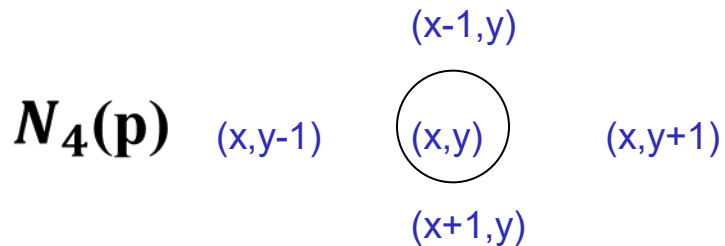
**FIGURE 2.31**

(a) Two sets of coordinates,  $A$  and  $B$ , in 2-D space. (b) The union of  $A$  and  $B$ . (c) The intersection of  $A$  and  $B$ . (d) The complement of  $A$ . (e) The difference between  $A$  and  $B$ . In (b)–(e) the shaded areas represent the member of the set operation indicated.

# Some Basic Relationships between Pixels

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## Neighbors of a pixel



# Adjacency

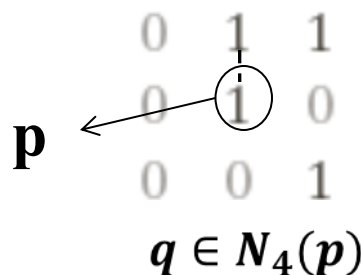
Adjacency is the relationship between two pixels  $p$  and  $q$

$V$  is a set of intensity values used to define adjacency

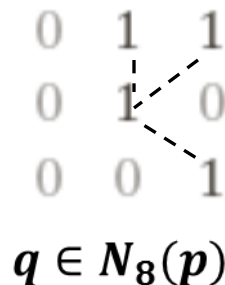
- Binary image:  $V=\{1\}$  or  $V=\{0\}$
- Gray level image:  $V \subseteq \{0, 1, \dots, 255\}$   
 $f(p) \in V$  and  $f(q) \in V \Rightarrow$  Intensity constraints

Three types of adjacency:

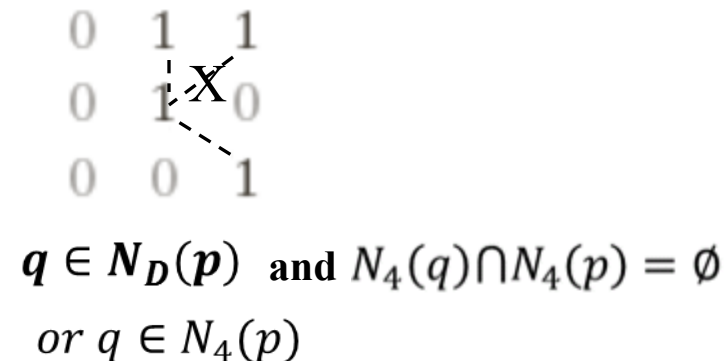
4-adjacency



8-adjacency



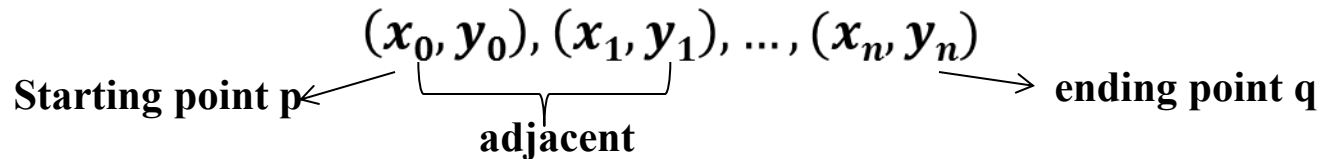
m-adjacency





# Connectivity

- Path from p to q: a sequence of distinct and adjacent pixels with coordinates



- **Closed path:** if the starting point is the same as the ending point
- p and q are *connected*: if there is a path from p to q in S
- **Connected component:** all the pixels in S connected to p
- **Connected set:** S has only one connected component

Are they connected sets?

```

0  1  1
0  1  0
0  0  1
  
```

```

0  1 - - 1
   |  \
0  1   0
   |  \
0  0   1
  
```

```

0  1 - - 1
   |  \
0  1   0
   |  \
0  0   1
  
```

# Regions

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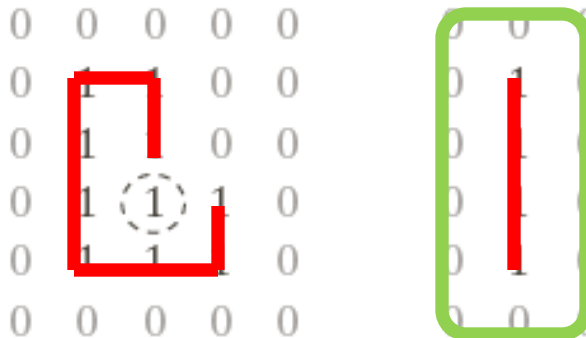
- $R$  is a region if  $R$  is a connected set
- $R_i$  and  $R_j$  are adjacent if  $R_i \cup R_j$  is a connected set

$$\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array} \left. \begin{array}{l} \\ \\ \\ \\ \end{array} \right\} \begin{array}{l} R_i \\ \\ \\ R_j \end{array}$$

# Boundaries

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- Inner boundary (boundary) -- the set of pixels each of which has at least one **background neighbor**
- Outer boundary – the boundary pixels in the background



# Distance Measures

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For pixels  $p$ ,  $q$ , and  $z$ , with coordinates  $(x,y)$ ,  $(s,t)$  and  $(v,w)$ ,  $D$  is a distance function or metric if

$$(a) \ D(p, q) \geq 0 \quad D(p, q) = 0 \text{ iff } p = q$$

$$(b) \ D(p, q) = D(q, p), \quad \text{and}$$

$$(c) \ D(p, z) \leq D(p, q) + D(q, z)$$

## Distance Measures

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**Euclidean distance**  $D_e(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$

**City-block (D4) distance**  $D_4(p, q) = |x - s| + |y - t|$

**Chessboard (D8) distance (Chebyshev distance)**

$$D_8(p, q) = \max(|x - s|, |y - t|)$$

# Distance: Sample Problem

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D4 distance

6

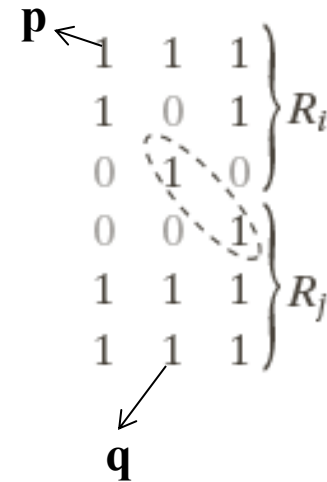
D8 distance

5

Euclidean distance

$$\sqrt{1 + 5^2}$$

Distance vs length of a path?



# Mathematic Tools

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## Array/Matrix operations

## Linear/nonlinear operations

**Linearity:**  $H[a_i f_i(x, y) + a_j f_j(x, y)] = a_i H[f_i(x, y)] + a_j H[f_j(x, y)]$

## Arithmetic Operations – single pixel operations

- Image averaging, image subtraction, image multiplication

## Set and logic operations

## Spatial operations

- Single pixel operations and neighborhood operations

## Image transformation

## Probabilistic methods

# Mathematic Tools

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## Array versus Matrix operations

### Array Multiplications

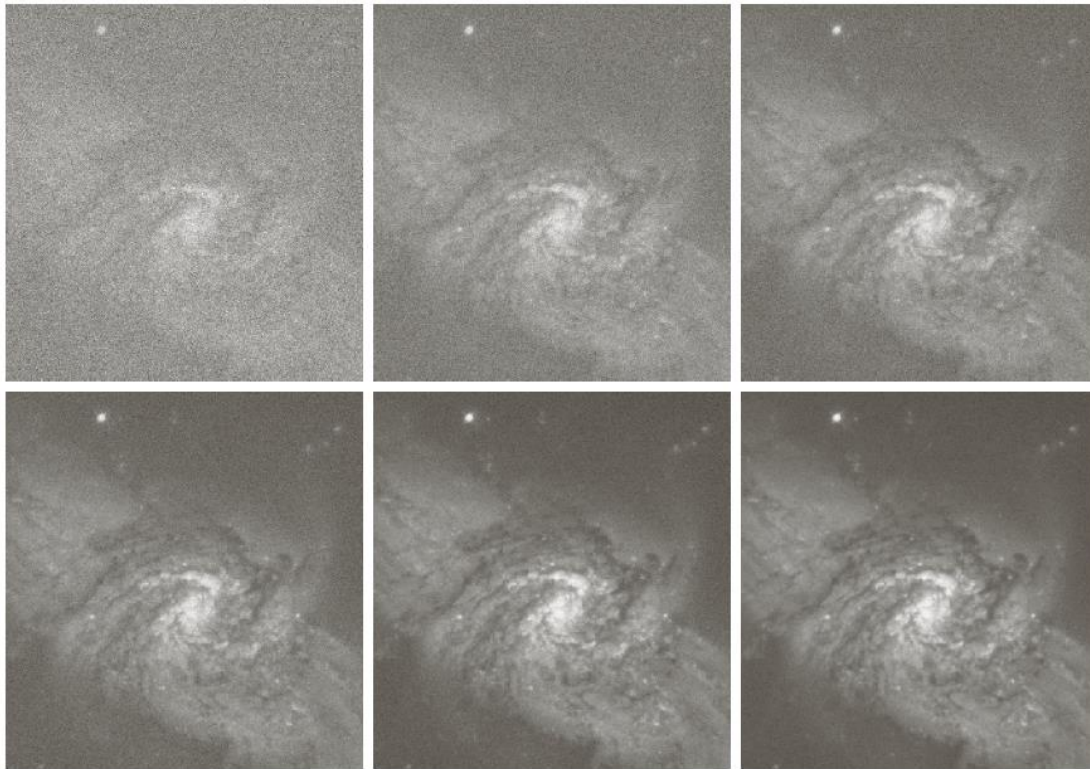
$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \cdot \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

### Matrix Multiplications

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$



# Image Averaging – Noise Reduction



a b c  
d e f

**FIGURE 2.26** (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)–(f) Results of averaging 5, 10, 20, 50, and 100 noisy images, respectively. (Original image courtesy of NASA.)

$$g(x, y) = f(x, y) + \eta(x, y)$$



$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$$

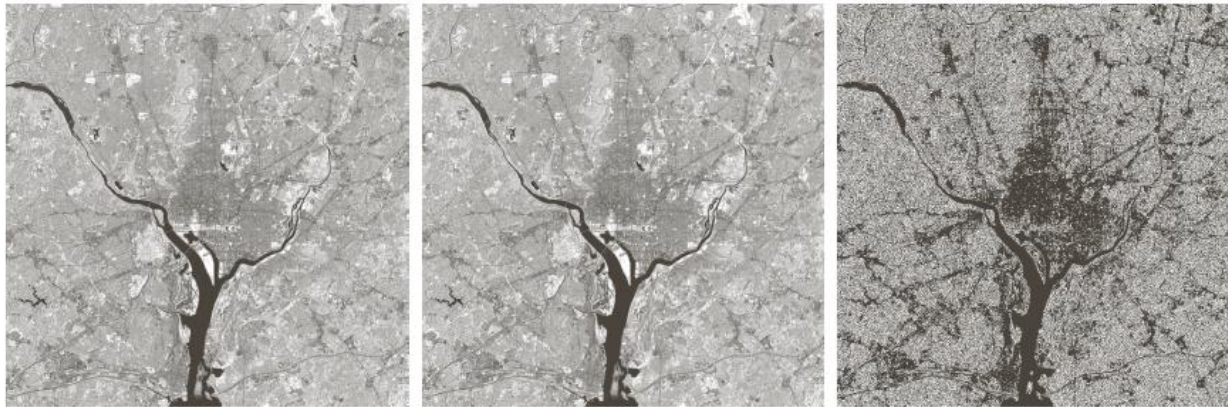
$$E\{\bar{g}(x, y)\} = f(x, y)$$

$$\sigma_{\bar{g}(x,y)}^2 = \frac{1}{K} \sigma_{\eta(x,y)}^2$$

**Assumption:** the noise is uncorrelated in image and has zero mean

# Image Subtraction – Enhance Difference

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a b c

**FIGURE 2.27** (a) Infrared image of the Washington, D.C. area. (b) Image obtained by setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range  $[0, 255]$  for clarity.